

5G RAN

QCI-based Resource Control Feature Parameter Description

Issue	Draft B
Date	2026-01-30



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Contents

1 Change History.....1

2 About This Document.....4

2.1 General Statements.....4

2.2 Features in This Document.....4

2.3 Differences.....5

3 Overview.....6

4 QCI-based Resource Control.....8

4.1 Principles.....8

4.1.1 Basic Resource Management.....8

4.1.1.1 Configuring QCI-Group-based Resource Groups.....9

4.1.1.2 Configuring QCI Groups.....13

4.1.2 Enhanced Resource Management.....14

4.1.2.1 RB-Usage-based Rate Policy.....14

4.1.2.2 RB-Usage-based UE Selection Policy.....15

4.1.2.3 User-Rate-based Dynamic QCI Resource Management.....16

4.1.3 Collaboration with Other Resource Control Functions.....16

4.1.3.1 NR DU Cell Resource Management Between Network Slices.....16

4.1.3.2 RFSP-Group-based RB Resource Control.....20

4.2 Network Analysis.....20

4.2.1 Benefits.....20

4.2.2 Impacts.....20

4.3 Requirements.....21

4.3.1 Licenses.....21

4.3.2 Functions.....21

4.3.2.1 Prerequisite Functions.....22

4.3.2.2 Mutually Exclusive Functions.....23

4.3.2.3 Function Impacts.....28

4.3.3 Hardware.....37

4.3.4 Others.....37

4.4 Operation and Maintenance.....38

4.4.1 Data Configuration.....38

4.4.1.1 Data Preparation.....38

4.4.1.2 Using MML Commands..... 40

4.4.1.3 Using the MAE-Deployment..... 41

4.4.2 Activation Verification.....41

4.4.3 Network Monitoring.....41

5 Parameters..... 42

6 Counters..... 44

7 Glossary..... 45

8 Reference Documents.....46

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Modified the mechanism of user-rate-based dynamic QCI resource management. For details, see: 4.1.2.3 User-Rate-based Dynamic QCI Resource Management 4.2.2 Impacts 4.4.1.1 Data Preparation 4.4.1.2 Using MML Commands	Added parameters: - NRDUCellQciBearer.DynResULTargetRate - NRDUCellQciBearer.DynResDLTargetRate	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Added descriptions of the impact relationships of user-rate-based dynamic QCI resource management with UL and DL Decoupling and independent SUL deployment. For details, see 4.3.2.3 Function Impacts.

Revised the hardware requirements for user-rate-based dynamic QCI resource management. For details, see 4.3.3 Hardware.

Added descriptions of the user-rate-based dynamic QCI resource management mechanism for an NR DU cell TRP. For details, see 4.1.2.3 User-Rate-based Dynamic QCI Resource Management.

Added descriptions of the impact relationships of user-rate-based dynamic QCI resource management with the following functions: downlink intra-band CA, downlink intra-FR inter-band CA, uplink intra-band CA, and uplink intra-FR inter-band CA. For details, see [4.3.2.3 Function Impacts](#).

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added user-rate-based dynamic QCI resource management. For details, see: • 2.2 Features in This Document • 4.1.2.3 User-Rate-based Dynamic QCI Resource Management • 4.2.2 Impacts • 4.3.2.1 Prerequisite Functions • 4.3.2.2 Mutually Exclusive Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the DYNAMIC_QCI_GROUP_SW option to the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between QCI-based resource control and user-rate-based dynamic network slice resource management. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added UE selection guarantee for QCI groups. For details, see: • 4.1.2.2 RB-Usage-based UE Selection Policy • 4.2.2 Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the QCI_GRP_UE_SEL_GURA_POL_SW option to the NRDUCellRes.SchedulingPolicySwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added a mutually exclusive relationship between QCI-based resource control and FSD transmission mode. For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between enhanced L4S congestion control based on rate adjustment and QCI-based resource control. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
None	None	4 QCI-based Resource Control

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
QCI-based resource control	User-rate-based dynamic QCI resource management is supported only in SA networking. Other functions have no differences between NSA and SA.	4 QCI-based Resource Control

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
QCI-based resource control	Supported only in low frequency bands	4 QCI-based Resource Control

3 Overview

As 5G technology advances, networks are extending their capabilities to deliver more refined services and differentiated experiences. 5G QoS Identifiers (5QIs) or QoS class identifiers (QCIs) are used to differentiate between common subscribers and premium subscribers, as well as their corresponding services. For details about 5QIs and QCIs, including their mapping, see the "SA Service-to-Bearer Mapping" and "NSA Service-to-Bearer Mapping" sections in *QoS Management*. In standalone (SA) networking, the 5G core network (5GC) sends a 5QI to the gNodeB over the NG interface, and the gNodeB then maps this 5QI to a locally configured QCI. In non-standalone (NSA) networking, the evolved packet core (EPC) sends a QCI to the eNodeB over the S1 interface, and the eNodeB then forwards the QCI to the gNodeB over the X2 interface.

To offer premium subscribers differentiated experiences while still ensuring the basic experience for common subscribers, the gNodeB is allowed to classify subscribers or UEs into different QCI groups. These groups are then assigned specific resource control policies, enabling more tailored services. [Figure 3-1](#) and [Figure 3-2](#) depict how 5QIs or QCIs are transmitted and mapped in different networking scenarios.

Figure 3-1 5QI or QCI transmission in SA networking

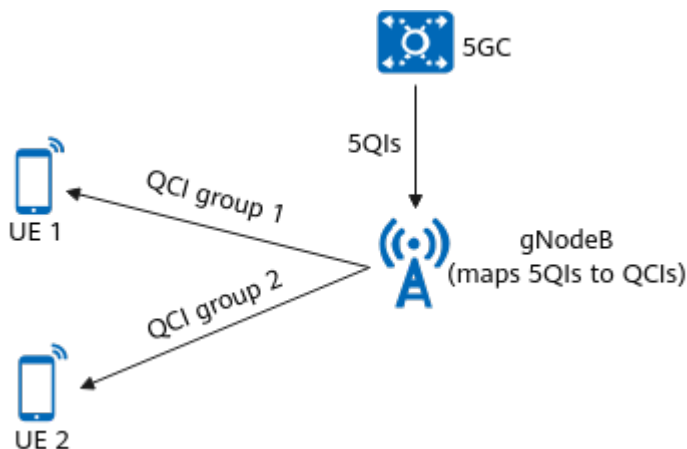
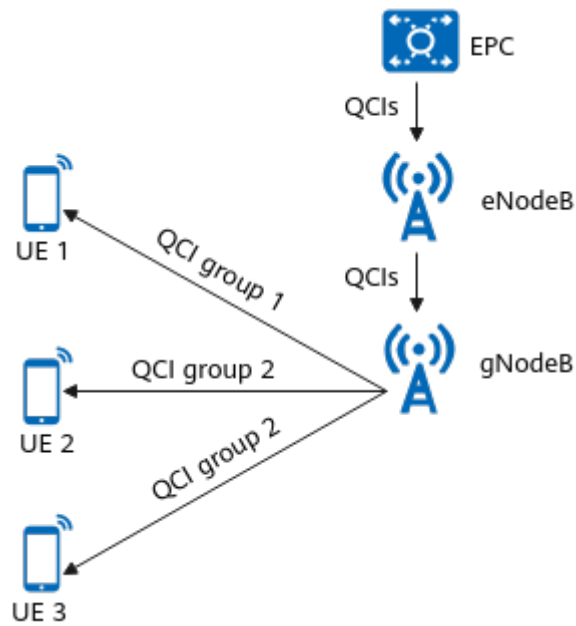


Figure 3-2 QCI transmission in NSA networking



NOTE

Common subscribers and premium subscribers are defined by operators.

4 QCI-based Resource Control

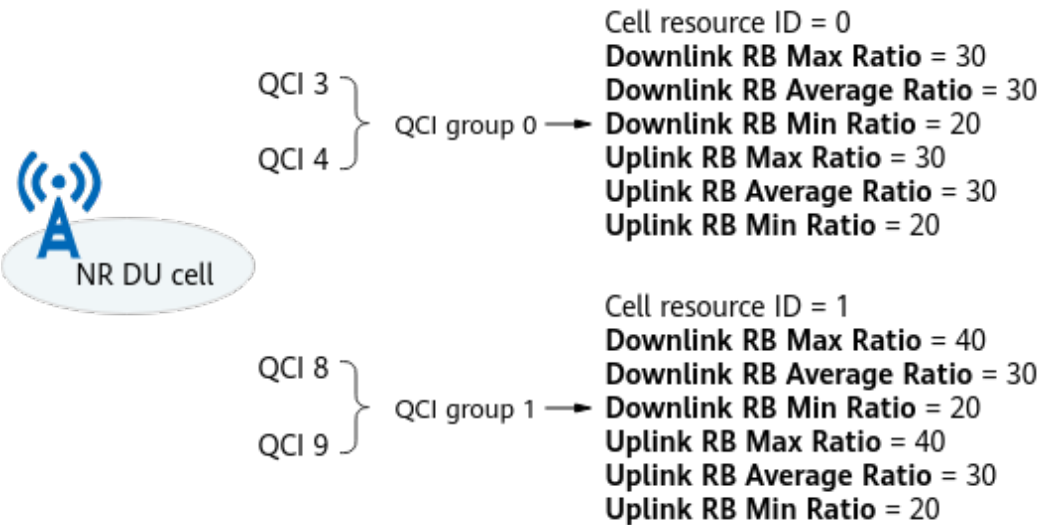
4.1 Principles

Premium subscribers configured with a high priority or a guaranteed bit rate (GBR) in a cell can preempt resource block (RB) resources, affecting the basic experience of common subscribers in the cell. To address this, QCI-based resource control is introduced, offering premium subscribers differentiated experiences while still ensuring the basic experience for common subscribers. This function manages RB resources based on QCI groups. The gNodeB is allowed to allocate RB resources to different QCI groups and dynamically manage the RB resources occupied by these groups, thereby ensuring service experience for UEs in each group. RB resources refer to physical downlink shared channel (PDSCH) or physical uplink shared channel (PUSCH) resources. This function is controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter.

4.1.1 Basic Resource Management

To enable QCI-based resource control, it is necessary to configure both QCI-group-based resource groups and QCI groups, as shown in [Figure 4-1](#).

Figure 4-1 Configuration example for QCI-based resource control



4.1.1.1 Configuring QCI-Group-based Resource Groups

QCI-group-based resource groups are configured using **NRDUCellRes** MOs with the **NRDUCellRes.ResUsageObject** parameter set to **QCI_GROUP**.

The configuration of a resource group takes effect only when it is bound to a QCI group. For instructions on how to bind a resource group to a QCI group, see [4.1.1.2 Configuring QCI Groups](#).

A cell supports up to 12 QCI-group-based resource groups. All QCIs associated with the same resource group must belong to the same operator. In multi-operator scenarios where an NR cell is shared among operators, one or more resource groups can be configured for each operator. Note that the total number of resource groups of all operators cannot exceed 12.

Related Parameters

To configure a resource group, it is necessary to specify the minimum, average, and maximum RB proportions, as described in [Table 4-1](#).

Table 4-1 Parameters related to resource groups for QCI-based resource control

Parameter Name	Parameter ID	Description
Downlink RB Max Ratio	NRDUCellRes.DIRbMaxRatio	Specify the maximum proportion of a cell's available PDSCH or PUSCH RBs that can be used by UEs in a resource group.
Uplink RB Max Ratio	NRDUCellRes.UIRbMaxRatio	Resources scheduled for retransmissions upon discontinuous transmissions (DTXs) are not restricted by the NRDUCellRes.DIRbMaxRatio or NRDUCellRes.UIRbMaxRatio parameter. If there are multiple resource groups, the maximum proportion of RBs available for UEs in a resource group is not restricted only by the configured maximum RB proportion. Instead, it is restricted by the smaller of either the maximum RB proportion or 100% minus the sum of the minimum proportions of RBs available for UEs in other resource groups.
Downlink RB Average Ratio	NRDUCellRes.DIRbAverageRatio	Specify the proportion of a cell's available PDSCH or PUSCH RBs that can be preferentially used by UEs in a resource group.
Uplink RB Average Ratio	NRDUCellRes.UIRbAverageRatio	
Downlink RB Min Ratio	NRDUCellRes.DIRbMinRatio	Specify the minimum proportion of a cell's available PDSCH or PUSCH RBs that can be used by UEs in a resource group. The RBs within the RB quota specified by either parameter are exclusively used by these UEs.
Uplink RB Min Ratio	NRDUCellRes.UIRbMinRatio	

Configuration Rules

The configuration rules for uplink and downlink RB proportions are identical. However, the preceding six related parameters cannot be set to **255** simultaneously.

For example, to configure the downlink RB proportion, set one or more of the following parameters based on the network plan: **Downlink RB Max Ratio**, **Downlink RB Average Ratio**, and **Downlink RB Min Ratio**. For their meanings, see [Table 4-1](#).

- If two of these parameters are set for a resource group, the following rules apply:
 - **Downlink RB Max Ratio** $\geq$ **Downlink RB Average Ratio**
 - **Downlink RB Average Ratio** $\geq$ **Downlink RB Min Ratio**

- **Downlink RB Max Ratio $\geq$ Downlink RB Min Ratio**
- If all of these parameters are set for a resource group, the following rule applies: **Downlink RB Max Ratio $\geq$ Downlink RB Average Ratio $\geq$ Downlink RB Min Ratio**.
- If multiple resource groups are configured for an NR DU cell, the sum of **Downlink RB Average Ratio** or **Downlink RB Min Ratio** set for all resource groups must not exceed 100%.

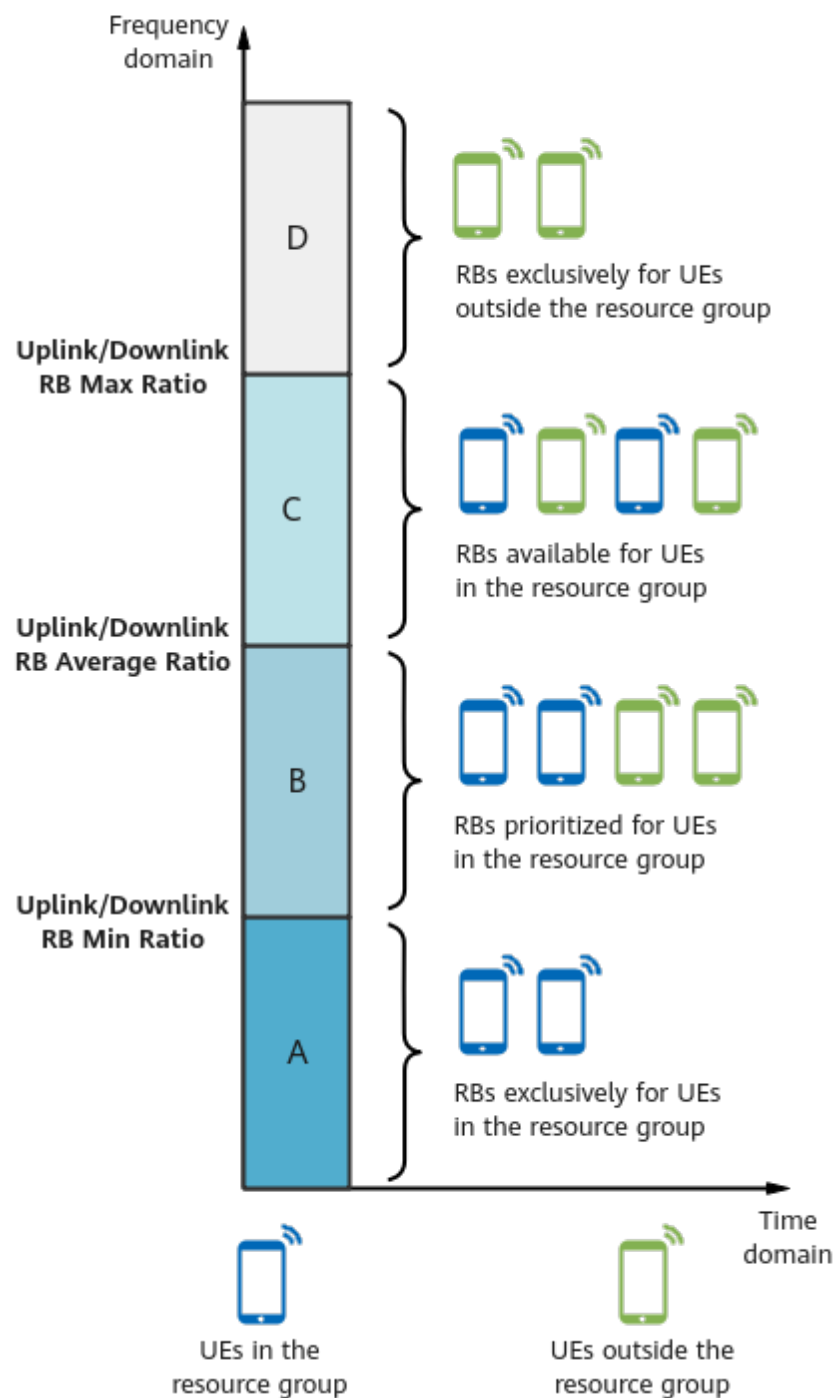
RB Consumption Rules

The resource quotas represented by **Uplink/Downlink RB Max Ratio**, **Uplink/Downlink RB Average Ratio**, and **Uplink/Downlink RB Min Ratio** (described in [Table 4-1](#)) set for a resource group are used as follows:

- RBs within quota **A** are exclusively used by UEs in the resource group.
- RBs within quota **B** are preferentially used by UEs in the resource group, and can be used by UEs outside the group when available.
- RBs within quota **C** are shared among all UEs, not preferentially used by UEs in the resource group.
- RBs within quota **D** are exclusively used by UEs outside the resource group.

The preceding example of RB configuration and consumption is shown in [Figure 4-2](#).

Figure 4-2 Example of RB configuration and consumption



In enhanced Mobile Broadband (eMBB) scenarios, the **Downlink RB Average Ratio** or **Uplink RB Average Ratio** parameter can be set to ensure the resource usage of QCI groups. You are not advised to set the **Downlink RB Min Ratio** and **Uplink RB Min Ratio** parameters, as this will affect the cell's RB usage.

The difference between the proportion of RBs used by UEs in each resource group in a cell and the configured proportion of RBs may exceed 1% when all of the

following conditions are met: (1) The sum of **Downlink RB Average Ratio** or **Uplink RB Average Ratio** set for all resource groups is 100%. (2) There are a sufficient number of UEs performing full buffer services in each group. (3) The values of **Downlink RB Average Ratio** or **Uplink RB Average Ratio** set for these groups differ greatly (by more than 70%).

When only one QCI-specific bearer is set up for a UE who initiates services, RBs are measured based on the configuration of the resource group associated with that QCI. When multiple QCI-specific bearers are set up for a UE, RBs are measured according to the rules described in [Table 4-2](#).

Table 4-2 Resource measurement rules when multiple QCI-specific bearers are set up for a UE

Scheduling Direction	Service Type	RB Measurement Rule
Uplink	Services with a specific QCI initiated by a UE ^a	The RBs used by services with this QCI are counted into the RBs for the resource group associated with this QCI.
	Services with different QCIs concurrently initiated by a UE	All the RBs used by the UE are counted into the RBs for the resource group corresponding to the bearer with the highest logical channel priority.
Downlink	Services with a specific QCI initiated by a UE	The RBs used by services with this QCI are counted into the RBs for the resource group associated with this QCI.
	Services with different QCIs concurrently initiated by a UE	RBs are measured based on different QCIs. ^b
a: The bearers carrying services with different QCIs must belong to different logical channel groups. For the related configurations, see the descriptions of QoS guarantee on the gNodeB side in the "QoS Guarantee for Uplink Scheduling" section in <i>QoS Management</i>. b: The RBs used by services with each QCI are counted into the RBs for the resource group associated with that QCI.		

4.1.1.2 Configuring QCI Groups

QCI groups are configured using **NRDUCellOpQci** MOs. A QCI specified by the **NRDUCellOpQci.Qci** parameter can be bound to a resource group specified by the **NRDUCellRes.CellResId** parameter by setting the **NRDUCellOpQci.QciGrpCellResId** and **NRDUCellRes.CellResId** parameters to the same value.

A QCI cannot be added to a QCI group when any of the following conditions is met:

- The QCI value is 1, 2, 5, 65, 66, 67, or 69.
- The **gNBQciBearer.ServiceType** parameter corresponding to the QCI is set to **URLLC_HIGH_RELIABILITY_SERVICE**.
- The **UL_GRANT_FREE_QCI_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter associated with the QCI is selected.
- The **gNBDUMacParamGroup.SchedulingType** parameter associated with the QCI is set to **VOICE_SCH**, **IMS_SIGNALING_SCH**, or **HIGH_RELIABILITY_SCH**.

4.1.2 Enhanced Resource Management

In addition to basic resource management, enhanced resource management is introduced to better ensure the user experience for common subscribers and balance resource allocation among UEs within a QCI group. This is achieved through the RB-usage-based rate policy, RB-usage-based UE selection policy, and user-rate-based dynamic QCI resource management.

To understand the meanings of the **Uplink/Downlink RB Max Ratio**, **Uplink/Downlink RB Average Ratio**, and **Uplink/Downlink RB Min Ratio** parameters mentioned in this section, see [Table 4-1](#).

4.1.2.1 RB-Usage-based Rate Policy

When both minimum rate guarantee for GBR services and minimum rate guarantee for non-GBR services take effect, a significant number of RB resources are preempted by GBR services and non-GBR services. This leaves fewer resources available for common subscribers, ultimately impacting their user experience negatively. To better ensure the basic experience for common subscribers, the RB-usage-based rate policy is introduced. This policy dynamically regulates the effectiveness of minimum rate guarantees for GBR and non-GBR services based on the **Uplink/Downlink RB Average Ratio** parameter set for a resource group, ensuring resource allocation for common subscribers. It does not apply to voice service UEs.

This policy is controlled by the **RB_USAGE_BASED_RATE_POL_SW** option of the **NRDUCellRes.SchedulingPolicySwitch** parameter, which can be selected only when the **NRDUCellRes.ResUsageObject** parameter is set to **QCI_GROUP**.

- If this option is deselected and the proportion of RBs used by UEs in a QCI group is less than or equal to the **Uplink/Downlink RB Max Ratio** parameter value, minimum rate guarantees for GBR and non-GBR services take effect.
- If this option is selected and the proportion of RBs used by UEs in a QCI group is less than or equal to the **Uplink/Downlink RB Average Ratio** parameter value, minimum rate guarantees for GBR and non-GBR services take effect. If this option is selected and the proportion of RBs used by UEs in a QCI group exceeds the **Uplink/Downlink RB Average Ratio** parameter value, minimum rate guarantees for GBR and non-GBR services do not take effect, and more RBs are allocated to common subscribers.

NOTE

For details about minimum rate guarantee for GBR services and minimum rate guarantee for non-GBR services, see *QoS Management*.

4.1.2.2 RB-Usage-based UE Selection Policy

The RB-usage-based UE selection policy is introduced to prevent some UEs in a QCI group from overusing resources, which negatively impacts the user experience of other UEs in the same QCI group. This policy is recommended when cell resources are insufficient and multiple UEs are performing services in a QCI group. In addition, you are advised to set the **Uplink/Downlink RB Max Ratio** parameter to protect the services of UEs outside the QCI group. This policy is controlled by the **NRDUCellRes.QciGrpRbCtrlUeSelPol** parameter.

- If this parameter is set to **OFF**, this policy does not take effect.
- If this parameter is set to **AUTO**, **ARP_BASED**, or **MCS_BASED**, this policy takes effect. When the proportion of RBs used by UEs in a QCI group exceeds the **Uplink/Downlink RB Average Ratio** parameter value, the gNodeB periodically selects a specific UE with RB priority from that group and removes its priority access to RBs within the quota represented by **Uplink/Downlink RB Average Ratio**. Conversely, when the proportion of RBs used by UEs in a QCI group does not exceed the **Uplink/Downlink RB Average Ratio** parameter value, the gNodeB periodically selects a specific UE without RB priority from that group and grants it priority access to RBs within the quota represented by **Uplink/Downlink RB Average Ratio**. Specific UEs are defined as follows:
 - If the parameter is set to **AUTO**, there are no specific UE restrictions.
 - If the parameter is set to **ARP_BASED**, specific UEs are those with an allocation and retention priority (ARP) value greater than the **NRDUCellRes.QciGrpRbCtrlArpThld** parameter value.
 - If the parameter is set to **MCS_BASED**, specific UEs are those with a modulation and coding scheme (MCS) index lower than the **NRDUCellRes.QciGrpRbCtrlMcsThld** parameter value.

Note that the **NRDUCellRes.QciGrpRbCtrlArpThld** parameter takes effect only when the **NRDUCellRes.QciGrpRbCtrlUeSelPol** parameter is set to **ARP_BASED**, and the **NRDUCellRes.QciGrpRbCtrlMcsThld** parameter takes effect only when the **NRDUCellRes.QciGrpRbCtrlUeSelPol** parameter is set to **MCS_BASED**.

When this policy takes effect, a smaller number of UEs in a QCI group increases the likelihood of a single UE repeatedly gaining and losing its RB priority, while a higher traffic volume of UEs in a QCI group increases the risk of service degradation for UEs outside the group. In such cases, you are advised to enable UE selection guarantee for QCI groups, which is controlled by the **QCI_GRP_UE_SEL_GURA_POL_SW** option of the **NRDUCellRes.SchedulingPolicySwitch** parameter.

- If this option is deselected, the function does not take effect. The gNodeB can remove RB priority from one UE and grant it to another UE within each UE selection period.
- If this option is selected, the function takes effect. The gNodeB removes or grants RB priority for only one UE within each UE selection period.

UE selection guarantee for QCI groups takes effect only when the RB-usage-based UE selection policy is in effect.

4.1.2.3 User-Rate-based Dynamic QCI Resource Management

Users have varying rate requirements for their services. If the **Average UL RB Ratio** or **Average DL RB Ratio** parameter is set to an inappropriate value, the target service rates cannot be achieved, compromising service quality for affected users. To address this, user-rate-based dynamic QCI resource management is introduced to dynamically adjust the **Uplink/Downlink RB Average Ratio** parameter value for QCI-group-based resource groups. This function is controlled by the **DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter and applies only in low frequency bands.

With this option selected, if the **Uplink RB Average Ratio** or **Downlink RB Average Ratio** parameter is configured for a QCI-group-based resource group, this function allows the gNodeB to dynamically adjust RB allocation based on the uplink or downlink target rate specific to dynamic resources for UEs in the QCI group, rather than preferentially allocating RBs based on the configured parameter value. The uplink and downlink target rates specific to dynamic resources are specified by the **NRDUCellQciBearer.DynResULTargetRate** and **NRDUCellQciBearer.DynResDLTargetRate** parameters, respectively. Specifically, if the average proportion of uplink or downlink RBs estimated by the gNodeB for all QCI groups in an NR DU cell transmission reception point (TRP) is less than the configured **Uplink RB Average Ratio** or **Downlink RB Average Ratio** parameter value, RBs are allocated based on the configured parameter value for all QCI groups in the NR DU cell TRP. For details about the NR DU cell TRP, see *Cell Management*.

If the **Uplink RB Min Ratio** or **Downlink RB Min Ratio** parameter is also configured for the QCI-group-based resource group, RBs within the quota represented by **Uplink/Downlink RB Min Ratio** can be preferentially used by UEs in the QCI group, and can be used by UEs outside the group when available.

4.1.3 Collaboration with Other Resource Control Functions

To support service-specific or UE-specific resource control on the radio access network (RAN) side, QCI-based resource control can be enabled together with NR DU cell resource management between network slices or RFSP-group-based RB resource control.

To understand the meanings of the **Downlink RB Max Ratio**, **Downlink RB Average Ratio**, and **Downlink RB Min Ratio** parameters mentioned in this section, see [Table 4-1](#).

NOTE

In SA networking, NR DU cell resource management between network slices can be enabled to control resources for UEs in different network slice groups. For details, see *Network Slicing*.

In NSA networking, RFSP-group-based RB resource control can be enabled to control resources for UEs in different RAT/Frequency Selection Priority (RFSP) groups. For details, see *Resource Control in NSA Networking*.

4.1.3.1 NR DU Cell Resource Management Between Network Slices

When both QCI-based resource control and NR DU cell resource management between network slices are enabled for a cell, the gNodeB performs resource

control based on the network slice group and QCI group to which a UE belongs. If a UE belongs only to a network slice group, see *Network Slicing* for its resource control policy. If a UE belongs only to a QCI group, see [RB Consumption Rules](#) for its resource control policy. If a UE belongs to both a network slice group and a QCI group, its resource control policy is determined by the network slice group's resource control policy, as described in [Collaboration with Mean-Type Network Slice Groups](#) and [Collaboration with TTI-based Network Slice Groups](#).

Collaboration with Mean-Type Network Slice Groups

[Table 4-3](#) describes the resource control policies when the network slice group to which a UE belongs uses the mean-type control policy.

Table 4-3 Resource control policy for a UE that belongs to both a mean-type network slice group and a QCI group

Single-UE Multi-Slice RB Management Policy ^a	Resource Control Granularity	Scheduling Direction	Activated Resource Control Function	Effective Configuration
Disabled	UE-specific	Uplink	NR DU cell resource management between network slices	Parameters configured for the network slice group
	UE-specific	Downlink	Both QCI-based resource control and NR DU cell resource management between network slices	• The larger of either the Downlink RB Min Ratio parameter set for the network slice group or that for the QCI group • The larger of either the Downlink RB Average Ratio parameter set for the network slice group or that for the QCI group • The smaller of either the Downlink RB Max Ratio parameter set for the network slice group or that for the QCI group

Single-UE Multi-Slice RB Management Policy ^a	Resource Control Granularity	Scheduling Direction	Activated Resource Control Function	Effective Configuration
Enabled	UE-specific	Uplink	NR DU cell resource management between network slices	Parameters configured for the network slice group
	Service-specific	Downlink	Both QCI-based resource control and NR DU cell resource management between network slices	• The larger of either the Downlink RB Min Ratio parameter set for the network slice group or that for the QCI group • The larger of either the Downlink RB Average Ratio parameter set for the network slice group or that for the QCI group • The smaller of either the Downlink RB Max Ratio parameter set for the network slice group or that for the QCI group
a: The single-UE multi-slice RB management policy is controlled by the SINGLEUE_MULTISLICE_RB_CTRL_SW option of the NRDUCellAlgoSwitch . <i>NetworkSliceAlgoSwitch</i> parameter, which affects the resource control granularity.				

Collaboration with TTI-based Network Slice Groups

Table 4-4 describes the resource control policies when the network slice group to which a UE belongs uses the TTI-based control policy.

Table 4-4 Resource control policy for a UE that belongs to both a TTI-based network slice group and a QCI group

Single-UE Multi-Slice RB Management Policy ^a	Resource Control Granularity	Scheduling Direction	Activated Resource Control Function	Effective Configuration
Disabled	UE-specific	Uplink	NR DU cell resource management between network slices	Parameters configured for the network slice group
	UE-specific	Downlink	Both QCI-based resource control and NR DU cell resource management between network slices	Parameters configured for the TTI-based network slice group. In addition, the base station determines whether to stop scheduling in the current TTI based on the proportions configured for the QCI group and the average PRB usage of UEs in the QCI group.
Enabled	UE-specific	Uplink	NR DU cell resource management between network slices	Parameters configured for the network slice group
	Service-specific	Downlink	Both QCI-based resource control and NR DU cell resource management between network slices	Parameters configured for the TTI-based network slice group. In addition, the base station determines whether to stop scheduling in the current TTI based on the proportions configured for the QCI group and the average PRB usage of UEs in the QCI group.
a: The single-UE multi-slice RB management policy is controlled by the SINGLEUE_MULTISLICE_RB_CTRL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter, which affects the resource control granularity.				

4.1.3.2 RFSP-Group-based RB Resource Control

When both RFSP-group-based RB resource control and QCI-based resource control are enabled for a cell, see *Resource Control in NSA Networking* for the resource control policy for a UE that belongs only to an RFSP group, and see [RB Consumption Rules](#) for the resource control policy for a UE that belongs only to a QCI group. For a UE that belongs to both an RFSP group and a QCI group, the gNodeB performs resource control based on both the RFSP group and QCI group, as described in [Table 4-5](#).

Table 4-5 Resource control policy for a UE that belongs to both an RFSP group and a QCI group

Scheduli ng Direction	Activated Resource Control Function	Effective Configuration
Uplink	RFSP-group-based RB resource control	Parameters configured for the RFSP group
Downlink	Both RFSP-group-based RB resource control and QCI-based resource control	• The larger of either the Downlink RB Min Ratio parameter set for the RFSP group or that for the QCI group • The larger of either the Downlink RB Average Ratio parameter set for the RFSP group or that for the QCI group • The smaller of either the Downlink RB Max Ratio parameter set for the RFSP group or that for the QCI group

4.2 Network Analysis

4.2.1 Benefits

This function enables resource control based on QCI groups to meet the requirements for differentiated network services.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

The network impacts of each function are as follows:

- RB-usage-based rate policy
If the **RB_USAGE_BASED_RATE_POL_SW** option of the **NRDUCellRes.SchedulingPolicySwitch** parameter is selected and the proportion of RBs used by a QCI group exceeds the configured average RB

proportion, minimum rate guarantees for GBR and non-GBR services do not apply to all QCIs in the QCI group.

- RB-usage-based UE selection policy

If the **NRDUCellRes.QciGrpRbCtrlUeSelPol** parameter is set to **AUTO**, **ARP_BASED**, or **MCS_BASED**, the uplink and downlink rates decrease for UEs without RB priority and increase for those with RB priority in a QCI group. After the **QCI_GRP_UE_SEL_GURA_POL_SW** option of the **NRDUCellRes.SchedulingPolicySwitch** parameter is selected, fewer UEs are selected within each UE selection period, resulting in slower rate changes across the group.

- User-rate-based dynamic QCI resource management

When the **DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter is selected and the resources required to achieve the uplink/downlink target rates specific to dynamic resources for UEs in a QCI group exceed the resources represented by **Uplink/Downlink RB Average Ratio**, the network impacts are as follows:

- If the target rate satisfaction level for the group is low, the user experience of UEs in this group improves.
- If the cell load is medium or heavy (PRB usage exceeding 30%) and there are a large number of cell edge UEs in the group, the uplink and downlink cell throughputs (**Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**) decrease.

The rate satisfaction level is calculated using the following formulas:

- For the uplink, assuming counters specific to a minimum GBR of 3 Mbit/s:
$$\text{Uplink rate satisfaction level} = (\text{N.Thp.UL.Flex.Index3} + \dots + \text{N.Thp.UL.Flex.Index14}) / (\text{N.Thp.UL.Flex.Index0} + \dots + \text{N.Thp.UL.Flex.Index14})$$
- For the downlink, assuming counters specific to a minimum GBR of 5 Mbit/s:
$$\text{Downlink rate satisfaction level} = (\text{N.Thp.DL.Flex.Index2} + \dots + \text{N.Thp.DL.Flex.Index14}) / (\text{N.Thp.DL.Flex.Index0} + \dots + \text{N.Thp.DL.Flex.Index14})$$

NOTE

Flexible performance counters (such as **N.Thp.UL.Flex.Index0**) can be configured using the **gNBFlexCounterObjCfg** MO. For details, see "Custom Object Type" in the "Object Type" section in *Performance Management*.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When the RB-usage-based UE selection policy takes effect (that is, the NRDUCellRes.QciGrpRbCtrlUeSelPolicy parameter is set to AUTO , ARP_BASED , or MCS_BASED), QCI-based resource control must be enabled.
Low-frequency TDD	QCI group	NRDUCellRes.ResUsageObject set to QCI_GROUP	<i>QCI-based Resource Control</i>	The RB-usage-based rate policy can be enabled only when the NRDUCellRes.ResUsageObject parameter is set to QCI_GROUP .
			<i>QCI-based Resource Control</i>	The NRDUCellRes.QciGrpRbCtrlUeSelPolicy parameter can be set to AUTO , ARP_BASED , or MCS_BASED only when the NRDUCellRes.ResUsageObject parameter is set to QCI_GROUP .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RB-usage-based UE selection policy	NRDUCellRes.QciGrpRbCtrlUeSelPol	<i>QCI-based Resource Control</i>	UE selection guarantee for QCI groups takes effect only when the RB-usage-based UE selection policy (specified by setting the NRDUCellRes.QciGrpRbCtrlUeSelPol parameter to AUTO , ARP_BASED , or MCS_BASED) is in effect.
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	User-rate-based dynamic QCI resource management (controlled by the DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw) parameter can be enabled only when the switch controlling QCI-based resource control is turned on.

4.3.2.2 Mutually Exclusive Functions

- Mutually exclusive functions of QCI-based resource control (controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Operator-specific RB management	RB_DYNAMI C_SHARING_ SW and MAX_RB_LI MIT_SW options of the NRDUCellAI goSwitch.Ra nSharingAlg oSwitch parameter	<i>Multi-Operator Sharing</i>
Low-frequency TDD	Operator-specific guarantee on the minimum number of available RBs	MIN_RB_GU ARANTEE_S W option of the NRDUCellAI goSwitch.Ra nSharingAlg oSwitch parameter	<i>Multi-Operator Sharing</i>
Low-frequency TDD	Dynamic RB control for UEs requiring low latency and high reliability	RB_DYNAMI C_CONTROL _SW option of the NRDUCellAI goSwitch.Hi ghReliability Switch parameter	<i>URLLC</i>
Low-frequency TDD	Cell Combination	NRDUCell.N rDuCellNet workingMod e set to HYPER_CELL _COMBINE_ MODE and NRDUCellM ultiTrp.Data TransMode set to BASIC_MOD E	<i>Cell Combination</i>

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>
Low-frequency TDD	Intra-operator preferential sharing of idle resources	NRDUCellNsMgmtConfig.NeSliceRfspGrpSharePool set to SAME_OPERATOR_PREFERRED	<i>Network Slicing Resource Control in NSA Networking</i>
Low-frequency TDD	FSD transmission mode	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL and NRDUCellMultiTrp.DataTransMode set to FSD_MODE	<i>Hyper Cell</i>
Low-frequency TDD	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORKSLICE_SW option of the NRDUCellCommonSw.ServiceDiffAlgoExtSwitch parameter	<i>Network Slicing</i>

- Mutually exclusive functions for the QCI (specified by the **NRDUCellOpQci.Qci** parameter) configured for a QCI group

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	High-reliability-oriented scheduling	gNBDUMacParamGroup.SchedulingType set to HIGH_RELIABILITY_SCH	<i>URLLC</i>
Low-frequency TDD	High-reliability services	gNBQciBearer.ServiceType set to URLLC_HIGH_RELIABILITY_SERVICE	<i>URLLC</i>
Low-frequency TDD	Voice service scheduling	gNBDUMacParamGroup.SchedulingType set to VOICE_SCH	None
Low-frequency TDD	IMS signaling scheduling	gNBDUMacParamGroup.SchedulingType set to IMS_SIGNALING_SCH	None
Low-frequency TDD	QCI-based UL Grant-free	UL_GRANT_FREE_QCI_S option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>URLLC</i>

- Mutually exclusive functions for QCI groups (with **NRDUCellRes.ResUsageObject** set to **QCI_GROUP**)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Intra-operator preferential sharing of idle resources	NRDUCellInsMgmtConfig.NetSliceRfspGrpSharePol set to SAME_OPERATOR_PREFERRED	<i>Network Slicing Resource Control in NSA Networking</i>

- Mutually exclusive functions of user-rate-based dynamic QCI resource management (controlled by the **DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTR OL_SW option of the NRDUCellAlgoSwitch . NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>
Low-frequency TDD	Service admission	QOS_FLOW_ADMISSION_SW option of the NRDUCellAlgoSwitch . RacAlgoSwitch parameter	<i>Admission Control</i>
Low-frequency TDD	RB-usage-based UE selection policy	NRDUCellRes.QciGrpRbCtrlUeSelPol set to AUTO , ARP_BASED , or MCS_BASED	<i>QCI-based Resource Control</i>

- Mutually exclusive functions of the RB-usage-based UE selection policy (specified by setting the **NRDUCellRes.QciGrpRbCtrlUeSelPol** parameter to **AUTO**, **ARP_BASED**, or **MCS_BASED**)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink resource isolation	DL_EMBB_PREFERENTIAL_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When QCI-based resource control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDifferenceExpSw parameter) is enabled together with downlink resource isolation and the NRDUCellRes.DIRbAverageRatio or NRDUCellRes.DIRbMinRatio parameter takes effect for FWA users in a QCI group, the downlink throughput of eMBB users decreases. If the NRDUCellRes.DIRbAverageRatio or NRDUCellRes.DIRbMinRatio parameter takes effect for non-FWA users in a QCI group, the average downlink throughput of FWA users will decrease more significantly.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	NRDUCellAIgoSwitch.UlDlDecouplingSwitch set to ON	<i>UL and DL Decoupling</i>	When QCI-based resource control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter) is enabled together with UL and DL Decoupling and is enabled in the SUL cell, the RB proportion configured for the TDD cell takes effect in the SUL cell.
Low-frequency TDD	Independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAIgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When QCI-based resource control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter) is enabled together with independent SUL deployment and is enabled in the SUL cell, the RB proportion configured for the TDD cell takes effect in the SUL cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Minimum rate guarantee for GBR services	None	<i>QoS Management</i>	When the RB-usage-based rate policy (controlled by the RB_USAGE_BASED_RATE_POL_SW option of the NRDUCellRes.SchedulingPolicySwitch parameter) takes effect, the minimum rate of GBR services may not be guaranteed.
Low-frequency TDD	Minimum rate guarantee for non-GBR services	Downlink: gNBDUMacParamGroup.DlGuaranteedRate Uplink: gNBDUMacParamGroup.UlGuaranteedRate	<i>QoS Management</i>	When the RB-usage-based rate policy (controlled by the RB_USAGE_BASED_RATE_POL_SW option of the NRDUCellRes.SchedulingPolicySwitch parameter) takes effect, the minimum rate of non-GBR services may not be guaranteed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellCfg parameter	<i>MIMO (TDD)</i>	When QCI-based resource control works with PDSCH or PUSCH MU-MIMO: - The proportion of RBs used by UEs in a QCI group equals the proportion of layers paired for these UEs. For example, if there are a total of six paired layers and four of them are paired for UEs in a group, UEs in the group use 4/6 of all available RBs. - Assume that NRDUCellRes.DIRbAverageRatio and NRDUCellRes.UIRbAverageRatio are configured. The proportion of RBs used by UEs in a QCI group can reach the configured proportion only when a sufficient number of UEs in the group perform full buffer services. In this case, the pairing success rate will decrease
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellCfg parameter		

RAT	Function Name	Function Switch	Reference	Description
				for the UEs that are outside the QCI group and highly correlated with UEs in the group. This will negatively impact user experience because both the average downlink cell and UE throughputs will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced L4S congestion control based on rate adjustment	- Enhanced uplink L4S congestion control based on rate adjustment: UL_L4SPLUS_FOR_FAIR_SW option of the NRDUCellUISch.UISchAlgoSwitch parameter - Enhanced downlink L4S congestion control based on rate adjustment: DL_L4SPLUS_FOR_FAIR_SW option of the NRDUCellDISch.DISchAlgoEnhSwitch parameter	<i>Adaptive Managed Latency</i>	When QCI-based resource control is enabled together with enhanced L4S congestion control based on rate adjustment, the gNodeB restricts the available RBs for L4S UEs based on the available resources of the QCI resource group associated with the UEs, which may cause the calculated maximum rate and fair rate to be inaccurate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	NRDUCellIgoSwitch.UlDlDecouplingSwitch set to ON	<i>UL and DL Decoupling</i>	When user-rate-based dynamic QCI resource management is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments. When user-rate-based dynamic QCI resource management is enabled together with UL and DL Decoupling for both an SUL cell and its associated TDD cell, the NRDUCellQciBearer.DynResUlTargetRate and NRDUCellQciBearer.DynResDlTargetRate parameter values configured for the TDD cell also apply to the SUL cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAIgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When user-rate-based dynamic QCI resource management is enabled for an SUL cell in independent deployment mode, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments. When user-rate-based dynamic QCI resource management is enabled together with independent SUL deployment for both an SUL cell and its associated TDD cell, the NRDUCellQciBearer.DynResULTargetRate and NRDUCellQciBearer.DynResDLTargetRate parameter values configured for the TDD cell also apply to the SUL cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResDlTargetRate parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResDlTargetRate parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResUlTargetRate parameter value configured for the PCell also applies to the SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When user-rate-based dynamic QCI resource management is enabled, the NRDUCellQciBearer.DynResUlTargetRate parameter value configured for the PCell also applies to the SCells.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For user-rate-based dynamic QCI resource management: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

For other functions:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

Core network requirements:

- In SA networking, the 5GC must configure dedicated 5QIs for services that require resource control.
- In NSA networking, the EPC must configure dedicated QCIs for services that require resource control.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-6 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-6 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
QCI-based Diff Service Experience Switch	NRDUCellFeatureSw.QciBasedDiffServExpSw	Select the RB_DYNAMIC_CONTROL_SW option of this parameter.
NR DU Cell ID	NRDUCellRes.NrDuCellId	Set this parameter based on the network plan.
Cell Resource ID	NRDUCellRes.CellResId	Set this parameter based on the network plan.
Resource Usage Object	NRDUCellRes.ResUsageObject	Set this parameter to QCI_GROUP .
Downlink RB Max Ratio	NRDUCellRes.DlRbMaxRatio	Set this parameter based on the network plan.
Downlink RB Average Ratio	NRDUCellRes.DlRbAverageRatio	Set this parameter based on the network plan.
Downlink RB Min Ratio	NRDUCellRes.DlRbMinRatio	Set this parameter based on the network plan.
Uplink RB Max Ratio	NRDUCellRes.UlRbMaxRatio	Set this parameter based on the network plan.
Uplink RB Average Ratio	NRDUCellRes.UlRbAverageRatio	Set this parameter based on the network plan.
Uplink RB Min Ratio	NRDUCellRes.UlRbMinRatio	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCellOpQci.NrDuCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Operator ID	NRDUCellOpQci.Operat orId	Set this parameter based on the network plan.
QoS Class Identifier	NRDUCellOpQci.Qci	Set this parameter based on the network plan.
QCI Group Cell Resource ID	NRDUCellOpQci.QciGrp CellResId	Set this parameter based on the network plan.
Scheduling Policy Switch	NRDUCellRes.Schedulin gPolicySwitch	Select the RB_USAGE_BASED_RATE_P OL_SW option when the RB-usage-based rate policy is required.
QCI Group RB Control UE Selection Policy	NRDUCellRes.QciGrpRb CtrlUeSelPol	Set this parameter based on the network plan.
QCI Group RB Control ARP Threshold	NRDUCellRes.QciGrpRb CtrlArpThld	Set this parameter based on the network plan. This parameter takes effect only when the NRDUCellRes.QciGrpRbCtrl UeSelPol parameter is set to ARP_BASED .
QCI Group RB Control MCS Threshold	NRDUCellRes.QciGrpRb CtrlMcsThld	Set this parameter based on the network plan. This parameter takes effect only when the NRDUCellRes.QciGrpRbCtrl UeSelPol parameter is set to MCS_BASED .
Scheduling Policy Switch	NRDUCellRes.Schedulin gPolicySwitch	Select the QCI_GRP_UE_SEL_GURA_PO L_SW option when UE selection guarantee for QCI groups is required.
QCI-based Diff Service Experience Switch	NRDUCellFeatureSw.Qc iBasedDiffServExpSw	Select the DYNAMIC_QCI_GROUP_SW option when user-rate-based dynamic QCI resource management is required.
Dynamic Resource Uplink Target Rate	NRDUCellQciBear- er.DynResUlTargetRate	Set this parameter based on the network plan.
Dynamic Resource Downlink Target Rate	NRDUCellQciBear- er.DynResDlTargetRate	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
MAC Parameter Group ID	NRDUCellQciBearer.MacParamGroupId	Configure extended QCIs based on the network plan. Standardized QCIs that have been created by default do not need to be configured.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Activation Command Examples

```
//Enabling dynamic RB control for an NR DU cell
MOD NRDUCELLFEATURESW: NrDuCellId=7, QciBasedDiffServExpSw=RB_DYNAMIC_CONTROL_SW-1;
//Configuring RB resources for the NR DU cell
ADD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
ResStrategyType=MEAN_ONLY, DirbMaxRatio=80, DirbAverageRatio=30, DirbMinRatio=255,
UlrBMaxRatio=80, UlrBAverageRatio=30, UlrBMinRatio=255;
//Configuring a QCI group for the NR DU cell and associating the group with the RB resources
ADD NRDUCELLQCI: NrDuCellId=7, OperatorId=0, Qci=3, QciGrpCellResId=0;
ADD NRDUCELLQCI: NrDuCellId=7, OperatorId=0, Qci=4, QciGrpCellResId=0;
//Configuring the RB-usage-based rate policy
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
SchedulingPolicySwitch=RB_USAGE_BASED_RATE_POL_SW-1;
//Setting the RB-usage-based UE selection policy to automatic
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
QciGrpRbCtrlUeSelPol=AUTO;
//Configuring UE selection guarantee for QCI groups
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
SchedulingPolicySwitch=QCI_GRP_UE_SEL_GURA_POL_SW-1;
//Setting the RB-usage-based UE selection policy to ARP-based
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
QciGrpRbCtrlUeSelPol=ARP_BASED, QciGrpRbCtrlArpThld=15;
//Setting the RB-usage-based UE selection policy to MCS-based
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
QciGrpRbCtrlUeSelPol=MCS_BASED, QciGrpRbCtrlMcsThld=0;
//Configuring user-rate-based dynamic QCI resource management
//Configuring the ID of the MAC parameter group referenced by a bearer in the NR DU cell (with the QCI
representing the one configured for UEs in the group and the uplink and downlink target rates set to 10000)
MOD NRDUCELLQCI: NrDuCellId=7, Qci=128, MacParamGroupId=9, DynResUlTargetRate=10000,
DynResDlTargetRate=10000;
//Turning on the switch for user-rate-based dynamic QCI resource management
MOD NRDUCELLFEATURESW: NrDuCellId=7, QciBasedDiffServExpSw=DYNAMIC_QCI_GROUP_SW-1;
```

Deactivation Command Examples

```
//Removing the association between the QCI group of the NR DU cell and the RB resources
RMV NRDUCELLQCI: NrDuCellId=7, OperatorId=0, Qci=3;
RMV NRDUCELLQCI: NrDuCellId=7, OperatorId=0, Qci=4;
//Disabling user-rate-based dynamic QCI resource management
MOD NRDUCELLFEATURESW: NrDuCellId=7, QciBasedDiffServExpSw=DYNAMIC_QCI_GROUP_SW-0;
```

```
//Disabling the RB-usage-based rate policy
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
SchedulingPolicySwitch=RB_USAGE_BASED_RATE_POL_SW-0;
//Disabling UE selection guarantee for QCI groups
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP,
SchedulingPolicySwitch=QCI_GRP_UE_SEL_GURA_POL_SW-0;
//Disabling the RB-usage-based UE selection policy
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=QCI_GROUP, QciGrpRbCtrlUeSelPol=OFF;
//Removing the RB resources configured for the NR DU cell
RMV NRDUCELLRES: NrDuCellId=7, CellResId=0;
//Disabling dynamic RB control for the NR DU cell
MOD NRDUCELLFEATURESW: NrDuCellId=7, QciBasedDiffServExpSw=RB_DYNAMIC_CONTROL_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Once QCI-based resource control is enabled in a network environment with stable services, it takes effect when the value of any of the following indicators changes:

- In NSA networking:
 - **N.Thp.DL.Samp.QCI.Index0** to **N.Thp.DL.Samp.QCI.Index14**
 - QCI-specific downlink rate = $(\text{N.ThpVol.DL.QCI} - \text{N.ThpVol.DL.LastSlot.QCI}) / \text{N.ThpTime.DL.RmvLastSlot.QCI}$
 - QCI-specific uplink rate = $(\text{N.ThpVolForRate.UL.QCI} - \text{N.ThpVolForRate.UL.SmallPkt.QCI}) / \text{N.ThpTimeForRate.UL.RmvSmallPkt.QCI}$
- In SA networking:
 - **N.Thp.DL.Samp.DuCell5QI.Index0** to **N.Thp.DL.Samp.DuCell5QI.Index14**
 - 5QI-specific downlink rate = $(\text{N.ThpVol.DL.DuCell5QI} - \text{N.ThpVol.DL.LastSlot.DuCell5QI}) / \text{N.ThpTime.DL.RmvLastSlot.DuCell5QI}$
 - 5QI-specific uplink rate = $(\text{N.ThpVolForRate.UL.DuCell5QI} - \text{N.ThpVolForRate.UL.SmallPkt.DuCell5QI}) / \text{N.ThpTimeForRate.UL.RmvSmallPkt.DuCell5QI}$
- Downlink PRB usage = $(\text{N.PR.B.DL.Used.Avg} / \text{N.PR.B.DL.Avail.Avg}) \times 100\%$
- Uplink PRB usage = $(\text{N.PR.B.UL.Used.Avg} / \text{N.PR.B.UL.Avail.Avg}) \times 100\%$

4.4.3 Network Monitoring

You can observe the performance counters listed in [4.4.2 Activation Verification](#) to monitor the service status of QCI groups in a cell after QCI-based resource control is enabled.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Multi-Operator Sharing*
- *URLLC*
- *Cell Combination*
- *Admission Control*
- *Network Slicing*
- *Performance Management*
- *Adaptive Managed Latency*
- *Resource Control in NSA Networking*
- *FWA*
- *QoS Management*
- *UL and DL Decoupling*
- *Super Uplink*
- *MIMO (TDD)*
- *Hyper Cell*
- *Cell Management*